

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech (EC) Examination****Nov - Dec 2016****EC 308 Antenna and Wave Propagation****Date: 29/11/2016****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the below questions. [07]**

- a. If an antenna draws 12 A current and radiates 4 kW, then what will be its radiation resistance? [01]
- b. Fill the blank by choosing proper option. "Steradian is a measurement unit of _____." [01]
 - A. Point angle
 - B. Linear angle
 - C. Plane angle
 - D. Solid angle
- c. What is mean by Aperture efficiency? [01]
- d. Define beam width and half power beam width with respect to antenna power pattern. [02]
- e. Explain Antenna fading in brief. [02]

Q – 2 Answer any two questions. [14]

- a. A source has a cosine radiation-intensity pattern, $U=U_m\cos\theta$. where U_m is maximum radiation intensity, $0\leq\theta\leq\pi/2$, $0\leq\phi\leq2\pi$. [07]
 - a) Find Directivity D
 - b) If maximum radiation intensity is half its value then find D
 - c) Consider sine radiation intensity pattern instead of cosine, then find D.
- b. For array of two point source find out maxima, minima and HPBW points when both have equal amplitude and equal phase. Also draw radiation pattern. [07]
- c. Design a Rhombic antenna (tilt, length of leg, height above ground) at frequency 18 GHz and $\beta=32$. [07]

Q - 3 Answer any two questions. [14]

- a. Prove that for half wavelength dipole antenna the radiation resistance is approximate 280 ohms. Also prove that $Z=73n^2$ where Z is impedance of folded dipole antenna. [07]
- b. Draw the structure Log periodic antenna. Explain its structure and designing steps in detail using mathematical equations. [07]

- c. If *Aryabhata* (first satellite of India , Launch date :19th April 1975) send the signal of 1GHz in free space, what is the value of spatial attenuation in db at a distance of 1 Km. [07]

SECTION – II

Q - 4 Answer the below questions. [07]

- a. A Thin dipole antenna is $\lambda/50$ long. If $R_l = 1.5 \Omega$ then find antenna efficiency. [01]
- b. Select proper option: The radiation resistance of current element is given by _____. [01]
 A. $R_r = 80\pi^2(dl/\lambda)^2$ B. $R_r = 20\pi^2(dl/\lambda)^2$
 C. $R_r = 10\pi^2(dl/\lambda)^2$ D. None
- c. What is mean by parasitic elements? [01]
- d. Derive the equation of free space propagation (Friis formula). [04]

Q - 5 Answer any two questions. [14]

- a. Design a 5 turn helical antenna, which at 400Mhz operates in the normal mode. The spacing between the turns is $\lambda/50$. It is desired that the antenna possesses circular Polarization. Determine the [07]
 a) Circumference of the helix
 b) Length of a single turn
 c) Overall length of entire helix
 d) Pitch angle
- b. Explain Cassegrain Feed System. [07]
- c. Derive $R = \frac{(n-1)L}{n \cos \theta - 1}$ for non metallic dielectric lens antenna. [07]

Q - 6 Answer any two questions. [14]

- a. Design a Rectangular Patch antenna at frequency 15 GHz, $\epsilon_r = 2.87$ and $h = 0.2\text{cm}$. [07]
- b. Design 3 element yagi uda antenna at starting frequency of FM broadcast range. [07]
- c. Write the equation of Scalar potential and vector potential for short dipole and derive the equation of electric field components and magnetic field components from that. [07]
